

Barriers to Use: Perspectives on Environmental Research Software

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Keywords: Quality-in-Use, ISO/IEC 25019:2023, Research Software, Empirical Software Engineering, Survey, User-Centered Design.

Abstract: Research software is essential for modeling complex Earth system interactions, yet poor software quality-in-use (QIU) often hinders its adoption. This study investigates QIU barriers in environmental research software through a survey of environmental scientists (n = 45) grounded in the ISO/IEC 25019:2023 QIU framework. The survey evaluated seven QIU subcharacteristics relevant to environmental research workflows. Our results indicate that the subcharacteristics of *Accessibility*, *Suitability*, and *Experience* represent the most significant barriers to adoption. These barriers are associated with steep learning curves, the need to learn new programming environments, and the reliance on additional tools to extend software capabilities. These findings, primarily reflecting R-based environmental research workflows, highlight recurring challenges faced by users and provide guidance for improving user-centered research software.

1 INTRODUCTION

Research software is critical for a wide range of scientific disciplines, including physics, mathematics, chemistry, and biology. These software tools allow researchers to analyze complex datasets and develop theories to explain real-world phenomena (Kelly and Sanders, 2008). A specific subset of research software is environmental research software, which models intricate interactions within the Earth system. The Earth system is classified as an E-type system (Lehman, 1979) because of its inherent complexity, evolving dynamics, and emergent properties, which result from the interactions that occur within the natural world.

Consequently, researchers face a growing need for usable, trustworthy, and suitable simulation software that enhances their ability to conduct effective and efficient research. Unfortunately, much of the existing software has been developed by scientists who lack formal training in software engineering (Hannay et al., 2009; Wilson et al., 2014; Kelly

and Sanders, 2008; Rampersad et al., 2017). Consequently, research software often suffers from non-intuitive user interfaces, insufficient documentation, and steep learning curves (Rampersad et al., 2017; Milewicz and Rodeghero, 2019), thus diminishing user-perceived software quality.

The concept of software quality-in-use (QIU) addresses these challenges by considering the user's perception of the software in its context of use (Bevan, 1999). Formally, the International Organization for Standardization defines QIU as “the extent to which the system or product, when it is used in a specified context of use, satisfies or exceeds stakeholders’ needs to achieve specified beneficial goals or outcomes” (ISO 25019:2023, E).

To improve the QIU of environmental research software, we investigate the QIU obstacles encountered by environmental software users. We examine the QIU subcharacteristics defined in the ISO/IEC 25019:2023 QIU standard (ISO 25019:2023, E) across multiple environmental research domains. Specifically, we designed a questionnaire to answer the following research question (RQ):

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RQ1: What QIU subcharacteristics are barriers to use for environmental research software end-users?

We posit that QIU subcharacteristics that are rated low are areas where the software fails to meet user needs and expectations. In contrast, QIU subcharacteristics that receive high ratings do meet user expectations. Understanding the QIU obstacles users face will help software engineers and developers focus on the most critical QIU issues, ultimately improving the end-user experience and advancing environmental research software design.

Building on our prior research (Hastings et al., 2026), we also investigate the relationship between QIU concerns identified through different feedback mediums. Specifically, we aim to answer:

RQ2: How do the dominant QIU concerns identified through survey responses compare with those identified through online web forums?

In our previous work, we web-scraped four user forums for reactive transport modeling software, one subclass of environmental research software, to identify QIU characteristics and subcharacteristics that create barriers for end-users. Our current study extends our previous work by comparing QIU challenges identified from questionnaires with those revealed through web scraped forum data. If the obstacles align, we can recommend web scraping as an efficient tool for identifying domain-specific QIU concerns. Conversely, if discrepancies arise, it may indicate that web scraping is a context-specific approach that complements, rather than replaces, direct user feedback.

By identifying the QIU subcharacteristics that pose barriers to software use, this work contributes to improving user-centered environmental research software development. More broadly, this study reframes software adoption challenges as issues of software QIU rather than purely technical limitations. Understanding these barriers is critical because environmental research increasingly relies on computational tools, and limitations in software accessibility, user experience, or suitability can influence who is able to effectively conduct computational research.

The main contributions of this study are:

- Developing a questionnaire aligned with the ISO/IEC 25019:2023 standard to evaluate the QIU of environmental research software.
- Empirically identifying the QIU subcharacteris-

tics that most strongly hinder environmental research software adoption.

This paper is divided into several sections. In Section 2, we review the background of quality standards. Section 3 presents related work and how our study is distinct from other studies. In Section 4, we present our study design. In Section 5, we present the results of our study and discuss their implications in Section 6. The threats to validity of our research are presented in Section 7. Finally, Section 8 concludes our paper, including directions to enhance software development practices.

Data Availability: All data and data pipelines are archived on Zenodo at the following hyperlink: <https://doi.org/10.5281/zenodo.15492847>.

2 BACKGROUND

Quality standards have evolved significantly since their inception in the late 1970s, beginning with McCall's Model in 1977, and continue to be actively developed today (Singh and Kannoja, 2013; Suman, 2014; ISO 25019:2023, E). Early standards like McCall's Model emphasized aspects such as maintainability, reliability, and functionality (Singh and Kannoja, 2013; Suman, 2014) but devoted minimal attention to usability. As software systems have grown in complexity, usability has emerged as a critical component of software quality. As such, the developers of newer standards have aimed to integrate usability considerations more comprehensively and extend the standards to include other quality aspects impacting software use and adoption (Bevan, 1999, 1995; ISO 25019:2023, E).

Despite these advancements, many quality standards still focus primarily on product quality and usability and allocate limited attention to the user perception within the software context of use. The ISO/IEC 25019:2023 QIU standard distinguishes itself by being the first quality standard exclusively addressing software quality from the perspective of the user (i.e., QIU) ISO 25019:2023 (E). The ISO/IEC 25019:2023 QIU standard emphasizes QIU aspects based on user perception in real-world contexts, making it particularly suitable for evaluating software across various domains.

One way the ISO/IEC 25019:2023 QIU standard emphasizes user perceived quality is by establishing a framework of guidelines for software engineers to promote the extent to which their products meet user expectations (ISO 25019:2023, E). Unlike traditional quality standards, ISO/IEC 25019:2023 supports ongoing evaluation of QIU within a defined context of

use. This allows software engineers to define QIU goals, assess whether outcomes are trusted and useful, and refine software based on user feedback. This user-centric method ensures that software quality aligns closely with end-user needs, which is crucial in domains where usability, accessibility, compliance, experience, and other quality aspects directly impact software's perceived usefulness.

The ISO/IEC 25019:2023 QIU standard focuses on three overarching characteristics: **Beneficialness**, **Freedom from Risk**, and **Acceptability**. Each characteristic is further subdivided into subcharacteristics: *Usability*, *Suitability*, *Accessibility*, *Freedom from Economic Risk*, *Freedom from Environmental and Societal Risk*, *Freedom from Health Risk*, *Freedom from Human Life Risk*, *Experience*, *Trustworthiness*, and *Compliance*. These characteristics and subcharacteristics collectively provide a comprehensive framework for assessing software QIU.

For this study, we focus on evaluating seven QIU subcharacteristics: *Usability*, *Suitability*, *Accessibility*, *Freedom from Economic Risk*, *Experience*, *Trustworthiness*, and *Compliance*. We chose these subcharacteristics based on their direct relevance to the selection and effective use of environmental research software. *Usability*, *Suitability*, and *Accessibility* capture critical aspects of user expectations, needs, and interactions with software. The *Freedom from Economic Risk* subcharacteristic addresses cost-related barriers. *Experience* reflects the overall accumulation of knowledge. *Trustworthiness* reflects the confidence the user has that the software does what it is supposed to do. Finally, *Compliance* ensures that the software adheres to ethical, legal, and reliability standards crucial for environmental research software.

By narrowing our focus to these seven subcharacteristics, this study aims to identify and address the most significant QIU barriers faced by environmental research software users. We omitted the QIU subcharacteristics of *Freedom from Environmental and Societal Risk*, *Freedom from Health Risk*, and *Freedom from Human Life Risk*, as these concerns are generally not applicable to the operational context of environmental research software.

Applying the ISO/IEC 25019:2023 QIU framework to environmental research software is particularly important because software quality in this domain is not only determined by technical correctness, but also by whether researchers can effectively learn, trust, interpret, and apply computational tools in real scientific workflows. Consequently, evaluating QIU in environmental software provides insights into how well these tools support scientific discovery and collaboration across diverse research disciplines.

3 RELATED WORK

Quality-in-use studies play a critical role in evaluating software, providing insight into how effectively it meets user needs. However, prior studies have focused on general usability or product quality (Sagar and Saha, 2017) rather than the broader QIU dimensions defined in ISO/IEC 25019:2023.

Sagar and Saha (2017) and Almazroi (2021) examined the domains where usability studies (one subcharacteristic of QIU) are most prevalent, identifying a focus on medical, web, and mobile applications. QIU studies in other domains, particularly research software, remain largely underexplored. This gap is especially evident in environmental research software, where domain-specific QIU studies are scarce and generalizable evaluations are lacking.

Adding to this gap, Kelly and Sanders (2008) interviewed research software developers and identified several reasons for this deficit. Notably, they found that domain scientists, who often serve as software developers for environmental research software, rarely conduct usability studies for their immediate user base. Usability evaluations are more common when software is used outside the primary application domain, leaving critical domain-specific challenges in QIU unaddressed (Kelly and Sanders, 2008).

Similar observations have been reported across scientific software domains. Hannay et al. (2009) and Wilson et al. (2014) found that many scientific software projects are developed by domain scientists with limited software engineering training, and that usability practices such as user-centered design and usability testing are rarely applied. Likewise, Segal (2007) characterizes many scientific programmers as “professional end-user developers” who primarily design tools for their own research needs rather than for broader user communities. Additionally, Anzt et al. (2020) report that research software projects often lack sustained funding and institutional support, limiting long-term maintenance and sustainability. These structural challenges further contribute to adoption and use barriers in domain-specific research software.

To address these gaps, our work contributes a generalizable QIU questionnaire aligned with the ISO/IEC 25019:2023 QIU standard. This questionnaire is designed to systematically evaluate and improve software QIU, fostering better alignment with end-user needs and expectations.

4 STUDY DESIGN

To address existing research gaps, we designed and distributed a questionnaire (IRB 2025-1913-EXEMPT) to environmental scientists to empirically evaluate research software QIU. This section outlines the study population, questionnaire design, pilot study, questionnaire distribution, and data analysis. We created and distributed the questionnaire following the software engineering survey guidelines established by Linåker et al. (2015).

4.1 Study Population

The study population consists of environmental research scientists who use environmental research software to model complex interactions within Earth systems. These scientists span diverse fields, including climate science, hydrology, soil science, and ecology. Their reliance on these software tools and their domain-specific expertise make them uniquely positioned to provide valuable insights into the QIU barriers of environmental research software. Understanding these barriers is crucial for improving the QIU of research software, ultimately enhancing the field of environmental research.

4.2 Questionnaire Design

We designed our questionnaire based on RQ1 outlined in Section 1. To inform our design, we reviewed prior studies that used questionnaires to evaluate QIU (Canchari et al., 2023; Lew et al., 2012; Pradanita and Rochimah, 2020). However, we found that the primary focus of these studies was on usability and user satisfaction and did not address the full range of QIU subcharacteristics defined in the ISO/IEC 25019:2023 standard. Consequently, we developed a new questionnaire to address the seven QIU subcharacteristics described in Section 2. We implemented the questionnaire in Qualtrics¹. Our questionnaire is divided into two main categories, as follows: **General Questions:** This section gathers information about the participants' backgrounds and software usage. The questions include multiple-choice (MC) and short-answer (SA) questions. Multiple choice questions provide an "Other" option for respondents to type in a response if the list of MC options does not apply to them. These questions were intended to contextualize QIU responses across different researcher backgrounds and software experiences. The questions included in this section are shown in Table 1.

¹<https://www.qualtrics.com/>

Quality-in-Use Assessment: This section assesses respondents' QIU experiences with the software they primarily use. Statements are organized according to the QIU subcharacteristics within each characteristic. Statements related to the **Beneficialness** and **Acceptability** subcharacteristics used an ordinal scale ranging from 0 (strongly disagree) to 5 (strongly agree), based on the respondent's personal experience. The *Freedom from Economic Risk* statements use a yes/no/unsure scale. We converted the responses into numeric values: "yes" = 1, "no" = 0, and "not sure" = 0.5. We instructed respondents to skip statements that did not apply to their software. The statements included in this section are shown in Table 2:

4.3 Pilot Study and Distribution

Prior to distributing the questionnaire, we conducted a pilot study with a subset of our study population consisting of environmental research scientists who use modeling software. The purpose of this pilot study was to evaluate the clarity and design of the questionnaire. Based on feedback from pilot study participants, we refined the wording of the consent form, as well as several questions and statements to improve clarity for the target population.

Following these revisions, the finalized questionnaire was distributed to the target population described in Section 4.1. The survey was distributed across multiple rounds between February 2025 and March 2026. We distributed the survey using a Qualtrics link shared through diverse channels to maximize reach. Distribution included postings to online networking boards, including three professional organization specific boards and LinkedIn. In addition, the questionnaire link was distributed via email to collaborators within our professional network. By leveraging these diverse distribution channels, we aimed to reach a broad sample of environmental research scientists actively using environmental research software.

To protect respondent privacy and confidentiality, the Qualtrics questionnaire was configured to be fully anonymous, and no personally identifiable information was collected.

4.4 Data Analysis

We calculated summary statistics and conducted non-parametric analyses on the survey results using R v. 4.4.2 (R Core Team, 2024). Summary statistics were calculated for the *general* and *Freedom from Risk* questions and statements to characterize respondent demographics (i.e., career stage, education level, and

Table 1: General Survey Questions.

ID	General Question	Type
G1	What career level do you identify with?	MC
G2	What is the highest level of education you have completed?	MC
G3	How many years have you been in environmental research?	MC
G4	What is the primary discipline of your research?	MC
G5	What environmental modeling software or library do you primarily use?	SA
G6	How long have you used this software or library?	MC
G7	How do you rate your proficiency with this software or library?	MC
G8	Why do you use this environmental software or library instead of a different one to fulfill your primary research objective?	MC
G9	Have you ever used or tried any other environmental modeling software or library to fulfill your primary research objective?	MC
G9a	Follow-up question if G9 is yes: What other environmental modeling software or libraries have you used?	SA
G9b	Follow-up question if G9 is yes: Why do you no longer use these other environmental software or libraries?	MC

Table 2: QIU Survey Statements by Characteristic and Subcharacteristic.

Characteristic	Subcharacteristic	ID	Statement
Beneficialness	<i>Usability</i>	U1	The software does everything I want it to do.
		U2	It is easy to load data into the software.
		U3	The software operates with little to no monitoring or extra tasks.
		U4	I can export the output to a local or networked file directory.
		U5	The software runs efficiently within expected timeframes.
		U6	I can easily adjust software parameters based on my needs.
		U7	The software interface is intuitive and easy to navigate.
		U8	The software provides clear error messages that help me troubleshoot issues.
	<i>Accessibility</i>	A1	I can use the software on multiple computers.
		A2	I had the computational resources available before acquiring the software (i.e., I did not have to buy new computers or servers).
		A3	The software has a graphical user interface.
		A4	I needed to learn a programming language to use the software.
		A5	The software includes accessibility features, such as keyboard shortcuts or compatibility with screen readers.
	<i>Suitability</i>	S1	The software output is understandable.
		S2	The software makes the graphs and tables I need to assess the results.
		S3	I have to make graphs and tables of results in a different software.
		S4	The software provides scientifically accurate results for my needs.
Acceptability	<i>Experience</i>	E1	The software documentation was sufficient for me to learn how to use the software.
		E2	Tutorials and workshops helped me learn the software.
		E3	Additional resources would improve my ability to use the software effectively.
		E4	Other people in my organization have used this software before and were able to teach me how to use it.
		E5	There is a user community outside my organization (i.e., user forums, meetings, email groups) where I can collaborate and ask questions to learn how to use the software.
		E6	The software functionality is easy to remember.
		E7	The software stops running unexpectedly.
	<i>Trustworthiness</i>	T1	The software performs the correct mathematical operations for my needs.
		T2	The software documentation clearly explains mathematical concepts used in the software.
		T3	I trust the software does what it is supposed to do.
		T4	The software consistently produces reproducible results when rerun with the same inputs.
		T5	The software protects the integrity and security of my data.
	<i>Compliance</i>	C1	I trust the software is licensed.
		C2	The software meets relevant industry guidelines.
Freedom from Risk	<i>Freedom from</i>	FF1	The software is free and open source.
	<i>Economic</i>	FF1a	Follow-up if FF1 is no: I had to buy multiple licenses so other people in my organization could use the software.
	<i>Risk</i>	FF2	I had to buy new equipment to use the software.

technical expertise), capture the diversity of environmental software tools in use, and determine the proportion of software causing financial impact to users.

To assess whether responses significantly differed

across groups, we conducted non-parametric analysis on the **Beneficialness** and **Acceptability** responses. We excluded responses with missing or incomplete data from pairwise and rank-based statistical analysis.

ses. However, we retained that data for descriptive statistics where applicable. To ensure higher scores consistently represent more favorable responses, we reverse-coded four survey statements using Eq. 1. The reverse-coded statements were A4, S3, E3, and E7.

$$\text{Reverse-coded score} = (\text{max Likert score} + 1) - \text{original score} \quad (1)$$

We identified “issue” statements across participant responses (Table 2). For each statement, we calculated the median Likert response across participants and used a threshold of median value < 3 to indicate potential QIU issues. This approach allowed us to identify the most prominent QIU concerns in environmental software as reported by users.

To evaluate the within-subject and population-level differences, we employed two non-parametric tests that evaluate the differences in perceived software QIU aspects. To assess whether respondents rated survey statements differently within-subject (i.e., across the 31 Likert-scale statements), we applied the Friedman test, a non-parametric test for repeated measures. We then explored pairwise differences between items using the Wilcoxon signed-rank test with Benjamini-Hochberg correction for multiple comparisons.

To complement this analysis, we examined between-statement differences across the entire population using the Kruskal-Wallis test, a non-parametric one-way ANOVA alternative. For post hoc comparison, we used Dunn’s test with adjusted p -values to identify statement pairs with significant differences in median responses across the overall study population.

5 RESULTS

In this section we provide the results of our study, including a description of our participant demographics and outcomes of our questionnaire.

5.1 Participant Demographics

We estimate the environmental research software user population to exceed 2,000 individuals. After several rounds of distribution, we received 45 completed responses. Our sample size is justified for large populations ($n > 2000$), where the required sample size is driven primarily by the desired precision (margin of error) rather than the population size itself. Following established survey guidelines, exploratory studies may accept wider margins of error (e.g., $\pm 10\%$ to $\pm 15\%$), which correspond to substantially smaller

sample sizes (Conroy, 2018). Based on these guidelines, a sample size of 45 responses is sufficient to achieve a margin of error of approximately $\pm 15\%$ for large populations, situating this study within accepted limits for exploratory survey research.

Respondents represented a broad range of career stages, educational levels, and technical skill sets (Table 3). Most participants reported working in ecology (53%), followed by hydrology (18%), biogeochemistry (7%), climate science (4%), and oceanography (4%). All other scientific disciplines were represented by less than 2% of the participant pool. Among respondents, 36% reported working as tenure-track professors and 80% indicated that they had obtained a PhD.

Participants also reported extensive research experience: 49% indicated they had worked in the research sciences for more than 20 years, 22% for 6–10 years, and 16% for 16–20 years. Respondents with 0–5 or 11–15 years of experience each comprised less than 7% of the participant pool. In addition, more than 70% of respondents reported their proficiency with the software they use as either “proficient and very comfortable” or “knowledgeable and somewhat comfortable,” indicating a high level of familiarity with the software applications used in their research workflows.

These demographic characteristics provide context for interpreting the software tools reported by participants and the motivations behind their selection, which are described in the following section.

5.2 Software Applications Used

Survey participants provided detailed information in the general questions regarding the software applications they primarily use, the reasons for selecting those tools, whether they had attempted to use alternative software for the same purposes, and the reasons they ultimately chose not to adopt those alternatives (Table 4).

The most commonly used software was R, representing 76% usage either as a standalone tool or in combination with other software applications. This usage includes R’s extensive ecosystem of packages (e.g., *vegan*, *BiodiversityR*) that support ecological and environmental data analysis. Other software applications used by participants included MODFLOW (an open-source groundwater modeling program developed by the U.S. Geological Survey), MATLAB, and Python along with its numerous scientific computing libraries.

Responses indicated that R was selected primarily because it is easy to learn and use, inexpensive to

Table 3: Summary table of study participant population demographics.

Attribute	Response	Count
Career Level	Tenure track professor	16
	Government research professional	10
	Postdoctoral researcher	6
	Non-tenure track professor	5
	Graduate student	4
	Industry research professional	3
	Undergraduate student	1
Education Level	PhD	36
	Masters	3
	Bachelors	5
	High School	1
Years in Research	>20	22
	16-20	7
	11-15	3
	6-10	10
	0-5	3
Primary Research Discipline	Ecology	24
	Hydrology	8
	Biogeochemistry	3
	Climate	2
	Oceanography	2
	Fisheries	1
	Hydrogeology	1
	Plant Science	1
	Water Treatment	1
	Environmental Politics and Policy	1
	Geography	1
Software Proficiency	Expert	8
	Proficient and very comfortable	17
	Knowledgeable and somewhat comfortable	15
	Intermediate and not comfortable	3
	Novice and still learning software basics	2

purchase, and is what their organization uses. However, more than half of survey participants also reported challenges associated with using environmental research software, including the need to learn a programming language, limited accessibility features (e.g., support for screen readers), and insufficient learning resources.

Finally, 57% of respondents indicated that they had previously used additional software for the same research purposes. These participants indicated they no longer use these software, or use them less frequently, because there is other software available, licensing for the other software was too expensive, that it didn't fulfill their research goals and needs, and that it was difficult to learn and use.

These findings provide context for interpreting the QIU responses analyzed in the following section, which examines patterns in user-reported issues across software applications using non-parametric analysis.

Table 4: Summary table of software applications used, reasons for use, and reasons other software is not used.

Attribute	Response	Count
Software Used	R packages	34
	MODFLOW	5
	Matlab packages	4
	Python packages	4
	ArcGIS Pro	2
	Government Database	1
	Hydrus 1D	1
	JAGS	1
	Julia	1
	MARK	1
	Minitab	1
	None	1
	Vortex	1
	PMx	1
	SAS	1
	SigmaPlot	1
	USGS OTIS	1
Reasons for Use	It's what my organization uses	14
	It was easy to learn	13
	It was inexpensive to purchase	18
	It is easy to use	13
	Other	26
Other Software Use	Yes	26
	No	19
Why not other software	Licensing too expensive	9
	Requires additional computational resources	1
	Difficult to learn	6
	Didn't fulfill project needs/goals	9
	Not familiar with software environment	2
	Difficult to use	6
	Other software available	12
	Other	10

5.3 Non-Parametric Analysis Outcomes

The results of our questionnaire suggest that users perceive the QIU subcharacteristics of *Accessibility*, *Suitability*, and *Experience* as the barriers to environmental research software use. Survey statements received a wide range of responses, with median response scores ranging from 2 to 5 (Figs. 1- 2). This result is evident when comparing the median responses for the broader QIU characteristics of **Beneficialness** and **Acceptability**. For these characteristics, the *Accessibility*, *Suitability*, and *Experience* subcharacteristics consistently had the lowest median scores among all QIU subcharacteristics (Fig. 1). Specifically, participants indicated that they had to learn a new programming language to use the software (A4), that they have to use a second software application to make figures and tables (S3), and that additional resources would have improved their ability to learn and use the software (E3).

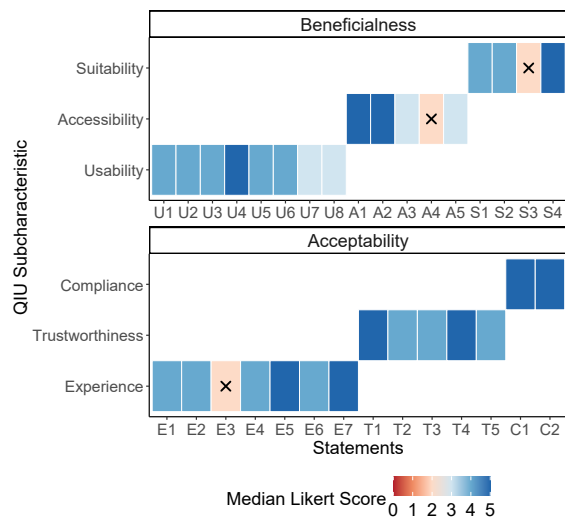


Figure 1: Matrix showing QIU statement median responses. Statements reflecting a QIU issue (i.e., median values < 3) are indicated with an 'X'. Reverse coding was applied to statements A4, S3, E3, and E7 so that higher scores reflect better outcomes.

To investigate differences in how participants rated QIU survey statements, we conducted a Friedman test. The Friedman test revealed statistically significant differences in how participants rated the QIU survey statements (U1 to C2; $\chi^2(30) = 152$, $p < 0.001$), indicating that participants did not answer all statements similarly. The Wilcoxon signed-rank post hoc tests identified 266 significant within-subject comparisons, out of 465 possible pairs (adjusted $p < 0.05$), suggesting variation in how users experience different QIU aspects (Fig. 2). These results indicate that respondents did not rate all QIU statements similarly and that perceptions of QIU varied across statements within individuals. This unique pattern demonstrates that user perceptions of different QIU aspects varied by their own experiences.

To complement the paired analysis, the Kruskal-Wallis test revealed significant differences in the overall median responses to statements (U1 to C2; $\chi^2(30) = 295.56$, $p < 0.001$). This result indicates that median response values significantly differed from statement to statement for the overall population. The Dunn test identified 238 significantly different comparisons out of 465 (adjusted $p < 0.05$), confirming that response medians were meaningfully different across survey statements. These findings suggest that participants experienced certain aspects of software QIU more positively or negatively than others, highlighting specific subcharacteristics as more problematic than others across the study population.

Regarding the *Freedom from Risk* subcharacter-

istic, more than 80% of respondents reported using free and open-source software (Fig. 3). Additionally, over 90% indicated they did not need to purchase additional hardware to use their primary software tools (Fig. 3). Out of the 45 participants, only one indicated they had to purchase additional licensing so other people in their organization could use the software. This result suggests minimal financial barriers associated with software use across environmental research scientists.

Together, the non-parametric analyses and median response distributions confirm that survey statements were not uniformly answered. In addition, visual inspection of median response values (Fig. 1) suggests that statements for the *Accessibility*, *Suitability*, and *Experience* subcharacteristics consistently received the lowest median ratings.

RQ1 Answer: The QIU subcharacteristics of *Accessibility*, *Suitability*, and *Experience* are the predominant barriers to environmental research software use, primarily in the context of R and its extensive package ecosystem. This means users had to learn a new programming language, multiple software applications are being utilized to analyze and interpret software outputs, and users are having difficulty learning the software.

6 DISCUSSION

In this section, we discuss the implications of our findings for environmental research software developers. We also compare the results of this survey with our previous work that investigated QIU barriers through web scraping of user forums.

6.1 Domain Implications

Our findings suggest that adoption of environmental research software is hindered primarily by challenges related to the QIU subcharacteristics of *Accessibility*, *Suitability*, and *Experience*. Survey responses indicated that users often need to learn new programming languages and invest considerable effort to understand and operate software applications. While AI tools can lower barriers to entry by assisting with code generation, it is imperative that users develop skills to evaluate and integrate these solutions into their workflows.

In the context of this paper, the widespread use of R among survey participants (76%) reflects broader trends in environmental sciences, where scripting lan-

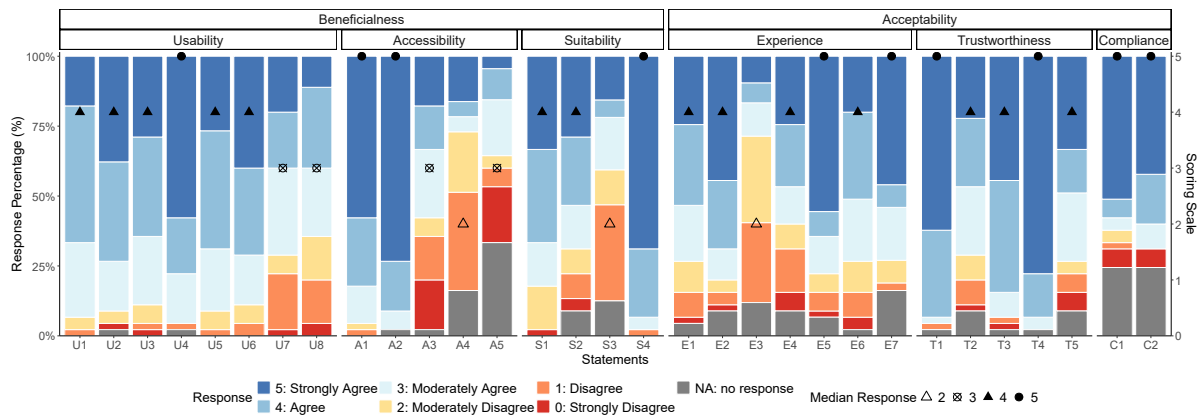


Figure 2: Response distributions and median values for Beneficialness and Acceptability QIU statements. The left-hand y-axis and stacked bars indicate the distribution of respondent agreement levels. Bar colors represent the level of respondent agreement, ranging from dark blue (“Strongly Agree”) to light blue (“Moderately Agree”), and from yellow (“Moderately Disagree”) to dark red (“Strongly Disagree”). The right-hand y-axis reflects the scoring scale (0-5) for each statement. Symbols indicate the median response value for each statement. Reverse coding was applied to statements A4, S3, E3, and E7 so that higher scores reflect better outcomes.

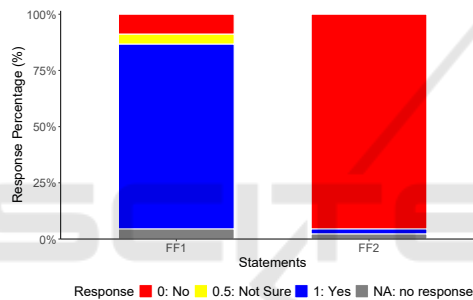


Figure 3: Response distribution for Freedom from Risk QIU statements. Bar colors represent respondent answers: red for “No”, yellow for “Not Sure”, blue for “Yes”, and gray for “NA”.

languages such as R and Python are widely adopted for data processing, statistical analysis, and reproducible workflows (Hampton, 2017). Indeed, these software applications are central to environmental research, supporting data-intensive modeling, visualization, and workflows across environmental domains, including ecological and climate monitoring (Hackenberger et al., 2025).

Together, these findings suggest that software adoption barriers in environmental research are not solely determined by the scientific capabilities of software applications. Rather, adoption is strongly influenced by how effectively software supports real-world research workflows and accommodates users with varying levels of computational expertise.

Accessibility: One key barrier identified in our survey is the requirement to learn new programming languages when adopting environmental research software. Survey participants also noted that

infrequent use of certain software applications makes them difficult to retain, with some users reporting that even short gaps in use can lead to loss of familiarity with the software.

This finding aligns with previous studies showing that many scientists who use research software lack formal training in programming or computational methods and instead acquire these skills informally through self-study or collaboration (Hannay et al., 2009; Wilson et al., 2014; Joppa et al., 2013). In many environmental science programs, emphasis is placed on field methods and theoretical foundations, while training in computational and data science skills is still emerging (Jones et al., 2022).

Taken together, these observations suggest that accessibility barriers in environmental research software are not only related to software availability, but also to the technical skills required to effectively use these tools over time.

Suitability: Participants reported that they frequently rely on secondary applications or R libraries to generate tables and visualizations necessary for interpreting and communicating their results. This fragmented workflow suggests that many environmental research tools do not fully support the analytical and reporting needs of users. Prior studies have emphasized the importance of integrating visualization capabilities directly into scientific software applications to improve the interpretability and suitability of model outputs (McInerney et al., 2014).

This finding suggests that suitability in environmental research software is closely tied to workflow completeness. Even when a software application performs its core analytical functions effectively, users

may still perceive it as insufficient if they must rely on additional software applications to interpret or communicate results.

Experience: Participants indicated that tutorials and workshops are essential resources for learning environmental software applications. However, frequent updates to software libraries can cause previously available capabilities to stop functioning, negatively affecting user experience and making it difficult for users to build long-term expertise with these applications. In addition, documentation is often incomplete or not updated to reflect version changes.

Previous studies have similarly observed that many environmental scientists rely on original publications describing early versions of software and may not fully understand how updated software applications operate (Joppa et al., 2013; Ahmed et al., 2015). For example, a survey conducted by Joppa et al. (2013) found that many domain scientists adopt software because it appeared in a peer-reviewed article, was recommended by colleagues, or was familiar from prior experience. Together, these findings suggest that user experience in environmental research software is influenced not only by the initial learning process, but also by the long-term stability of software documentation, the availability of learning resources, and the social pathways through which researchers discover and adopt software tools.

Beyond these QIU barriers, participants also emphasized the importance of free and open-source software in their adoption decisions. Many respondents indicated that open-source tools encourage continued use because they are freely available and widely supported within the research community. However, reliance on open-source platforms also raises sustainability concerns. As discussed in Section 2, Anzt et al. (2020) reported that many research software projects lack long-term funding, which limits their sustainability and maintainability.

This highlights an additional challenge to environmental research software adoption. That is, software applications that are most attractive to researchers because they are open and widely shared may also face challenges related to long-term maintenance, documentation, and technical support. This lack of sustainability ultimately hinders user experience over time as software applications become outdated or unmaintained.

Overall, our findings suggest several opportunities for improving environmental research software. First, improvements to training resources and documentation could help users overcome *Accessibility* and *Experience* barriers. Second, integrating visualization capabilities directly into software applications

and scripting libraries could improve the *Suitability* of software applications. Finally, improving documentation practices and software testing before new releases could enhance user *Experience* and increase long-term trust in environmental research software.

6.2 Results Compared to Previous Work

The results from this survey were used to evaluate whether web scraping of user forums can serve as an efficient method for identifying QIU-related software adoption barriers (Hastings et al., 2026). These studies are complementary rather than directly comparable because forum-mining captures spontaneously reported frustrations, whereas surveys capture prompted self-reported perceptions. Because the studies rely on different measurement units (forum-thread frequencies versus participant Likert responses), we qualitatively compared the dominant QIU subcharacteristics identified by each method to evaluate thematic convergence.

In our previous work, adoption barriers for reactive transport modeling software were primarily associated with *Usability* and *Suitability*. In the present survey study, the dominant barriers were associated with *Accessibility*, *Suitability*, and *Experience*. While *Usability* was not identified as a dominant barrier in the survey responses, *Suitability* emerged consistently across both studies, suggesting users experience recurring challenges in obtaining the expected capabilities and workflows needed for their research goals. The dominant QIU subcharacteristics identified across both studies are summarized in Table 5.

Table 5: Dominant QIU subcharacteristics identified across the two study methods. *Suitability* (bolded) represents a consistent barrier between the compared studies.

Study Method	Dominant QIU Subcharacteristics
Forum mining	Usability, Suitability
Survey	Accessibility, Suitability , Experience

Taken together, these findings suggest that different data collection methods reveal complementary aspects of software QIU. Survey instruments provide structured insights into user perceptions across predefined QIU dimensions, whereas web scraping captures informal questions and frustrations that emerge during real-world software use. Combining survey methods with web-scraped user feedback may therefore provide a more comprehensive understanding of software adoption barriers and end-user software challenges.

This recommendation addresses our second research question described in Section 1.

7 THREATS TO VALIDITY

In this section, we discuss potential threats to the validity of our study and the steps taken to mitigate them.

Construct and Internal Validity: Survey questions may fail to accurately capture the intended concepts if they are unclear or interpreted inconsistently. Because the study relies on self-reported data, responses may also be influenced by recall bias or stronger opinions from participants who chose to respond. To mitigate these risks, we followed established survey design guidelines by Linåker et al. (2015) and conducted a pilot study to evaluate question clarity, relevance, and alignment with the research objectives. Based on pilot study feedback, we refined the wording of the consent form and survey questions prior to distribution.

External and Conclusion Validity: The findings primarily reflect the perspectives of environmental research software users and may not fully generalize to other scientific domains. Sampling bias is also possible because the survey was distributed through professional networks and online communities rather than randomized sampling. Additionally, the study is based on a relatively small sample size ($n = 45$), with most respondents representing ecological disciplines. To mitigate these limitations, we distributed the survey across multiple environmental science communities and platforms (Section 4.3). Furthermore, the identified QIU barriers are consistent with findings from our prior web scraping study, suggesting these concerns may be more broadly experienced within the environmental research software community.

8 CONCLUSION

This study investigated the QIU of environmental research software by examining the barriers that end users face when adopting and using these tools. Using the ISO/IEC 25019:2023 QIU framework, we designed a questionnaire to evaluate seven QIU subcharacteristics: *Usability*, *Suitability*, *Accessibility*, *Freedom from Economic Risk*, *Experience*, *Trustworthiness*, and *Compliance*.

Our results indicate that users are most affected by challenges related to *Accessibility*, *Suitability*, and *Experience*. These findings suggest that barriers to software adoption are not solely related to scientific functionality, but also to how well software supports user workflows, learning processes, and interpretation of results. Improving the QIU of environmental research software therefore requires greater attention to

user-centered design, training resources, documentation quality, and integrated analytical capabilities.

We also compared the findings from this survey to those from a web-scraping study of user forums. Accordingly, we recommend that survey instruments and web scraping be used together to obtain a broader understanding of QIU concerns across research domains.

By applying the ISO/IEC QIU framework to environmental research software, this study contributes to broader efforts to improve the usability and effectiveness of scientific software tools. Ultimately, treating QIU as a first-class citizen of scientific software development can help ensure that computational tools better support the diverse needs of researchers and enable more effective scientific discovery.

ACKNOWLEDGMENTS

The authors acknowledge the use of ChatGPT to assist in shortening the Threats to Validity section text.

We thank the Montana State University Software Engineering and Cybersecurity Laboratory for their support of this work.

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